# RAHUL RAVI VK

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## SUMMARY


Instrumentation and controls engineer with 6+ years deploying closed-loop control, PLC/DCS automation, and sensor integration on live industrial plants, now pursuing an M.Eng. in Robotics. Combines hands-on hardware experience (PID tuning, field instrumentation, safety interlocks) with graduate work in ROS 2, perception, and motion planning. Seeking roles that bridge physical automation, instrumentation, and robotic systems.


## EDUCATION

**M.Eng. in Robotics** | University of Maryland, College Park — Sep 2025 – May 2027

GPA: 3.9 / 4.0 | Coursework: Planning and Perception for Autonomous Robots, Control Systems for Robotics, Autonomous Robotics, Robot Modeling, Intro to C++

**B.Eng. in Instrumentation & Control** | L.D. College of Engineering, Gujarat, India — Aug 2014 – May 2018

GPA: 3.3 / 4.0

## WORK EXPERIENCE

**Instrumentation & Controls Engineer** | *KRIBHCO Fertilizers Ltd, Gujarat, India* — Oct 2019 – Jul 2025

- Designed and tuned closed-loop control on pressure, flow, and level instrumentation across **Honeywell Experion PKS C300 DCS** and **Siemens PLCs**; cut loop overshoot, settling time, and process downtime by **30%** via systematic PID retuning and actuator nonlinearity analysis.
- Built advanced control strategies including pump torque-converter speed control and bistable-relay emulation logic, reducing equipment downtime by up to **50%** and false trips by **90%**.
- Owned full automation lifecycle for safety-critical units: field wiring, sensor calibration, PLC sequencing and interlocks, alarm rationalization, loop checking, FAT/SAT, and operator handover.

## PROJECTS

**Baron Mobile Robot — Hardware-Up Skid-Steer Platform** — *Raspberry Pi 4, Arduino, BNO055 IMU, Python*

- Built a skid-steer robot end-to-end from bare hardware: wheel-encoder odometry (10 CPR, 120:1 gearing, 65 mm wheels), **closed-loop PID velocity control**, and BNO055 IMU fusion over serial UART; implemented slip-compensated pivoting, teleop, and CSV data logging for controller identification.

**Vision-Based Arrow Detection for Autonomous Navigation** — *PyTorch, OpenCV, Raspberry Pi*

- Developed arrow-direction classifier for the Baron platform, progressing from classical CV (HSV, PCA) to a **ResNet18** transfer-learning model achieving **100% test accuracy**; deployed inference on Raspberry Pi 4 with live camera feed closing the loop with the motion controller.

**Optimal Control for Nonholonomic Mobile Robot (ENPM 667)** — *MATLAB, Genetic Algorithms, Adaptive PID*

- Reproduced an IEEE Access (2025) paper separating kinematic and dynamic control: GA-optimized kinematic gains $(K_\rho, K_\alpha, K_\beta)$ over a 6×12 grid of initial postures with closed-loop stability constraints, 2D cubic-spline interpolation, and an **adaptive PID** dynamic controller with gradient-descent gain tuning; verified Lyapunov stability of the adaptive law.

**Franka Panda Pick-and-Place** — *ROS 2 Humble, MoveIt 2, C++*

- Built end-to-end pick-and-place pipeline for a 7-DOF Franka arm using MoveIt 2 motion planning, a custom C++ execution node with named-pose sequencing, collision-aware planning, and gripper control.

**UR5 Browser-to-Robot Drawing System** — *ROS 2, IKPy, WebSockets*

- Designed decoupled architecture bridging a browser drawing interface to a Python IK backend (IKPy) via WebSockets, driving a UR5 in Gazebo with real-time Cartesian-to-joint-space tracking.

**Truck Twin-Trailer Simulation & Digital Twin** — *ROS 2 Control, Gazebo, SolidWorks, Blender*

- Modeled a multi-link articulated truck in SolidWorks, exported to URDF for ROS 2 Control simulation and FBX for Falcon Editor digital-twin deployment; implemented Python proportional controller for point-to-point navigation with LiDAR integration.

## TECHNICAL SKILLS

- **Industrial Automation & Instrumentation:** Honeywell Experion PKS C300 DCS, Siemens PLC (Ladder Logic), SCADA, safety interlocks, alarm rationalization, loop checking, FAT/SAT, field instrumentation, sensor calibration, 4–20 mA / HART
- **Control & Estimation:** PID, Adaptive PID, LQR, LQG, Kalman filtering, state-space modeling, Lyapunov stability, trajectory optimization, closed-loop system identification
- **Robotics & Simulation:** ROS 2 (Humble; Actions, Services, Topics, ros2_control), MoveIt 2, Gazebo, URDF/Xacro, RViz2, Falcon Editor, digital twins
- **Embedded & Hardware:** Raspberry Pi 4, Arduino Nano, BNO055 IMU, wheel encoders, GPIO, UART/serial, soldering, wiring, benchtop debugging
- **Programming & Tools:** C++17, Python (NumPy, SciPy, OpenCV, PyTorch, IKPy, ResNet / transfer learning, MATLAB/Simulink, Bash, WebSockets, SolidWorks, Blender, Linux (Ubuntu), Git, Docker

## CERTIFICATIONS

Siemens – Basics of PLC & SCADA | Honeywell – Experion PKS C300 Fundamentals